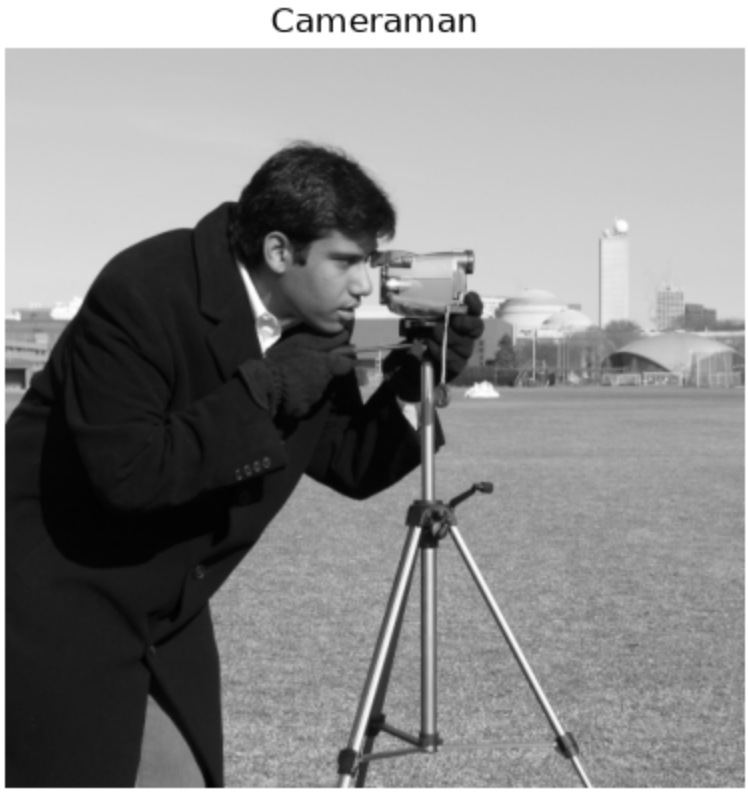
Anas Alhamzi - 2240003869

In [ ]:
```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('images/cameraman.tif')
plt.figure()
plt.imshow(img, cmap='gray')
plt.title("Cameraman")
plt.axis('off')
plt.show()
```



In [ ]:
```
from PIL import Image
import matplotlib.pyplot as plt

img2_gray = Image.open('images/lena_gray_256.tif').convert('L')

plt.figure()
plt.imshow(img2_gray, cmap='gray')
plt.title("Lena (Grayscale)")
plt.axis('off')
plt.show()
```



In [ ]:
```
cv2.imwrite('new_image.jpg', img)
```

Out[ ]: True

In [ ]:
```
print(img.shape)
print(img)
import numpy as np
img_array = np.array(img2)
print(img_array.shape)
print(img_array)
```

```
(512, 512, 3)
[[[200 200 200]
  [200 200 200]
  [200 200 200]
  ...
  [189 189 189]
  [190 190 190]
  [190 190 190]]

 [[200 200 200]
  [199 199 199]
  [199 199 199]
  ...
  [190 190 190]
  [190 190 190]
  [190 190 190]]

 [[199 199 199]
  [199 199 199]
  [199 199 199]
  ...
  [190 190 190]
  [190 190 190]
  [190 190 190]]

 ...

 [[ 25  25  25]
  [ 25  25  25]
  [ 27  27  27]
  ...
  [139 139 139]
  [122 122 122]
  [147 147 147]]

 [[ 25  25  25]
  [ 25  25  25]
  [ 26  26  26]
  ...
  [158 158 158]
  [141 141 141]
  [168 168 168]]

 [[ 25  25  25]
  [ 25  25  25]
  [ 27  27  27]
  ...
  [151 151 151]
  [152 152 152]
  [149 149 149]]]
(512, 512, 3)
[[[226 137 125]
  [226 137 125]
  [223 137 133]
  ...
  [230 148 122]
```

```
 [221 130 110]
 [200  99  90]]

[[226 137 125]
 [226 137 125]
 [223 137 133]
 ...
 [230 148 122]
 [221 130 110]
 [200  99  90]]

[[226 137 125]
 [226 137 125]
 [223 137 133]
 ...
 [230 148 122]
 [221 130 110]
 [200  99  90]]

...

[[ 84  18  60]
 [ 84  18  60]
 [ 92  27  58]
 ...
 [173  73  84]
 [172  68  76]
 [177  62  79]]

[[ 82  22  57]
 [ 82  22  57]
 [ 96  32  62]
 ...
 [179  70  79]
 [181  71  81]
 [185  74  81]]

[[ 82  22  57]
 [ 82  22  57]
 [ 96  32  62]
 ...
 [179  70  79]
 [181  71  81]
 [185  74  81]]]
```

In [ ]: